

Block-Structured Quad Meshing of Turbomachinery Surfaces

A cross-field guided domain-partition pipeline

Domain-Partition 3D

July 10, 2026

Goal & Pipeline

Goal: turn a raw turbine hub/shroud surface into a *block-structured, all-quad* mesh suitable for CFD.

Pipeline:

- 1 Unwrap cylinder $\rightarrow$ 2D
- 2 Cross-field (quad orientation)
- 3 Singularities
- 4 Streamlines / separatrices
- 5 Simplify & partition into blocks
- 6 Fill blocks with a structured grid



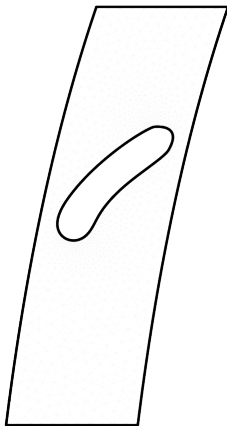
T1_9 turbine hub (cylindrical surface)

Step 1 — Unwrap the Cylinder

$$\theta = \text{atan2}(y, x), \quad s = r \theta, \quad t = z$$

- Constant radius $\Rightarrow$ **isometric** (distortion-free)
- 2D triangulation identical to 3D — only coordinates change
- Sheared strip with a blade-shaped hole; the two θ -walls are **periodic** (one blade pitch apart)

Code: `dp3d/unwrap_surface.py`

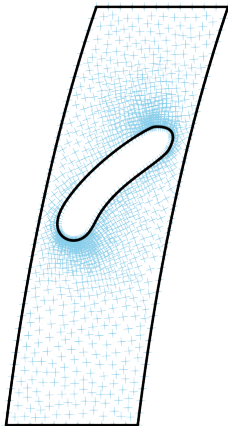


Step 2 — Cross-Field (Quad Orientation)

$$c(x) = (\cos 4\theta(x), \sin 4\theta(x))$$

- **4-RoSy** field: four directions per point, invariant under 90° rotation — exactly a quad's axes
- Smoothed by a harmonic solve, aligned to the walls
- A *representative vector* picks one of the four to integrate along

Knöppel et al., “Globally Optimal Direction Fields”, SIGGRAPH 2013

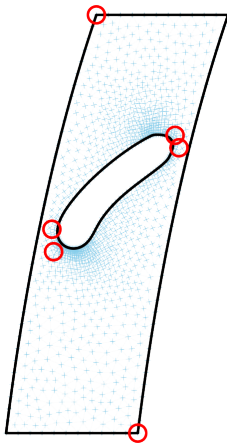


Step 3 — Singularities

$$\text{index}(p) = \frac{1}{2\pi} \oint_{\partial B(p)} d\theta, \quad \sum_i \text{index}(p_i) = \chi(\Omega)$$

- Points where a regular quad grid cannot exist (valence $\neq 4$)
- Here: the blade tips — irregular vertices are unavoidable
- **Poincaré–Hopf** fixes the total index budget

Detection: `dp3d/field/singularity_detector.py`; Kowalski et al. 2015

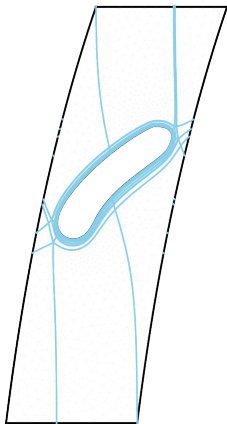


Step 4 — Streamline Integration

$$\mathbf{p}_{n+1} = \mathbf{p}_n + \frac{1}{2}(\mathbf{k}_1 + \mathbf{k}_2), \quad \mathbf{k}_1 = h\mathbf{v}(\mathbf{p}_n), \quad \mathbf{k}_2 = h\mathbf{v}(\mathbf{p}_n + \mathbf{k}_1)$$

- Trace field lines (*separatrices*) out of the singularities
- 2nd-order Runge–Kutta–Heun; stable near singular points
- Stop on a wall, another singularity, or a corner

StreamlineGenerator_v2; Kowalski 2015 (Prop. 9: index
+1 $\rightarrow$ 3 arms, $-1 \rightarrow$ 5 arms)

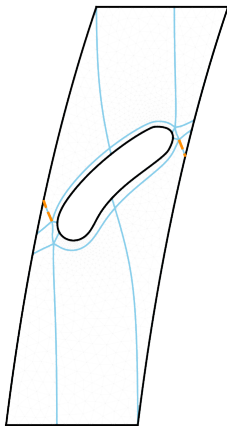


Step 5 — Simplify: Merge & Split

$$\gamma_{12}(s) = (1 - s) \gamma_1(s) + s \gamma_2(s)$$

- **Merge** near-duplicate arms at a singularity (angle criterion), blended by the formula above
- **Split** at transversal crossings $\rightarrow$ T-junctions
- Result: clean, quad-only block boundaries

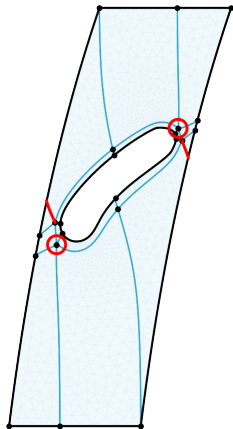
StreamlineMerging, StreamlineIntersectionSplitter; Xiao et al. 2020 (Alg. 2)



Step 6 — Blocks & Periodic Seam

- Extract the quad **blocks** bounded by the curves
- Collapse degenerate 3-sided seam wedges
- **Master–slave** seam: the right wall is the exact +pitch copy of the left — perfectly periodic
- Hanging junctions become T-nodes (variant T-a) or are continued into the domain (T-b)

`collapse_seam_wedges, symmetrize_seam_junctions`; Sauer
& Morsbach (DLR) 2023, sec. 2.7

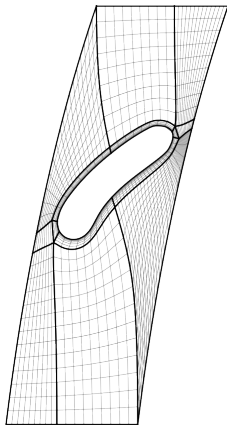


Step 7 — Fill Blocks: Transfinite Interpolation

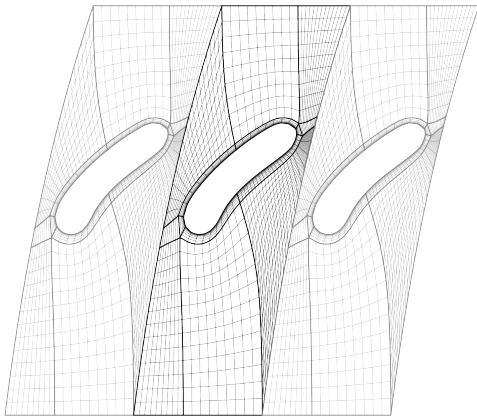
$$\begin{aligned} X(u, v) = & (1 - v)S(u) + v N(u) + (1 - u)W(v) + u E(v) \\ & - [(1 - u)(1 - v)c_{00} + u(1 - v)c_{10} \\ & + (1 - u)v c_{01} + uv c_{11}] \end{aligned}$$

- **Coons patch:** a structured grid from 4 block edges
- Conforming cell counts $\Rightarrow$ grids match across edges
- Wall clustering for the blade boundary layer

tfi_fill; Coons 1967

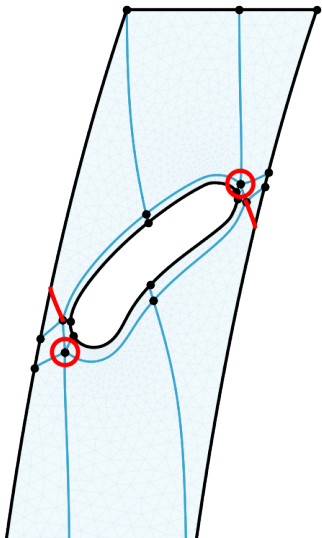


Periodicity Check — Tiled Passages

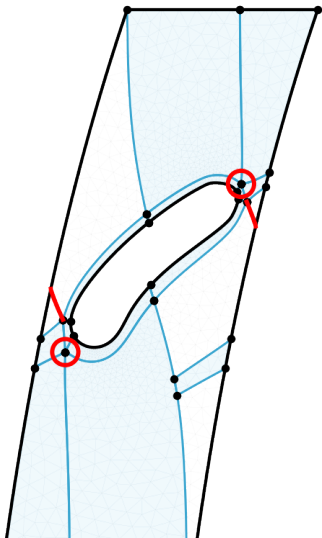


Three passages ($-\text{pitch}$, 0 , $+\text{pitch}$): grid points coincide across the interior seams — CFD-conforming periodicity

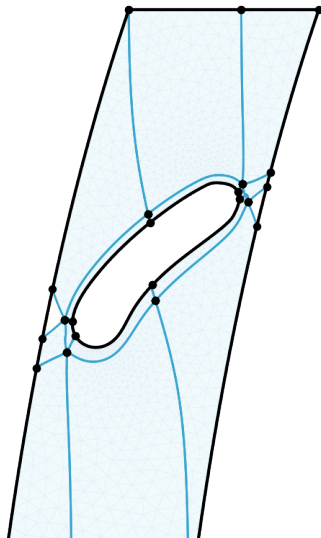
Method Comparison



Domain-Partition 3D



Domain Partition



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- Fully automatic: raw surface $\rightarrow$ block-structured quad mesh
- Cross-field $\rightarrow$ singularities $\rightarrow$ separatrices $\rightarrow$ blocks $\rightarrow$ TFI grid
- Periodic seam handled by master-slave conformity
- Hub works end-to-end; shroud + boundary-layer extraction in progress

- F. Knöppel, K. Crane, U. Pinkall, P. Schröder. *Globally Optimal Direction Fields*. ACM SIGGRAPH 2013.
- J. Xiao et al. *Robust topology decomposition of 2D domains*. 2020 (Alg. 2).
- N. Kowalski et al. *Block-structured quad meshing / PL-domain blocking*. 2015.
- T. Sauer, C. Morsbach (DLR). *Automated blocking for turbomachinery*. 2023, sec. 2.7.
- S. A. Coons. *Surfaces for Computer-Aided Design of Space Forms*. 1967.
- M. Reberol et al. *Quasi-structured quadrilateral meshing in Gmsh*.
- J. F. Thomas, J. F. Middlecoff. *Elliptic grid generation with control functions*. 1980.

Appendix — Labeled Diagnostics

